

How it's made: The Milky Way

Abstract:

Scientists have recognised galaxies for decades, but understanding how they were formed was another question entirely, based on components of the universe. I have researched multiple theories and studies on the formation of galaxies and have come to a solid theory on their formation from nebula and the mysterious Black Holes that cause all neighbouring stars and solar systems to orbit. This is all based on the Big Bang theory, where the universe is ever expanding from a central point which are the conditions scientists have speculated for the theory on the formation of Galaxies as the expansion of matter and dark matter leads to a cooling of the Universe thus allowing clusters of gas to form in space, eventually forming early-stage Galaxies.

Introduction:

The formation of galaxies had puzzled for centuries as they weren't even discovered until 1924 with the Hubble telescope (1) capturing the Andromeda Galaxy for the first time, which was groundbreaking and opened a rabbit hole of questions for physicists including how galaxies were formed.

As of 2024 multiple studies on the formation of galaxies by respected organisations and scientists have come with interesting articles and publications that have led us closer to truly understanding our own existence and pushing the limits of astrophysics and even quantum sciences with the understanding of minute particles and energy of atomic matter forming the galaxies around a central point where matter is the most dense, in theory having a tremendous gravitational pull.

Main Body:

There are 3 important factors and components of galaxies to discuss their formation, The Big Bang Theory, Classification and characteristics of galaxies and common threads, and research on the Hubble hinting at Galaxy evolution. We must base our theories on the most current theory of the creation of the universe. LaRocco and Rothstein (astrophysicists) release a publication in 2020 (2) which estimates the universe was created officially by the expansion or 'bang' of hot dense energy that grew "in the miniscule fractions of a second" and formed quarks and other fundamental particles such as electrons which began fusing into hydrogen, deuterium (heavy hydrogen atoms) and helium molecules due to the "continuously expanding" distance from the epicentre of the explosion condensing, discovered by Lemaître, 1931 (3). Over the next millions of years, was a rapid cooling of the universe, from roughly "100 nonillion Kelvin to 1 Billion Kelvin", according to Space.com (4) these molecules started to fuse with their energy forming 'Nebulas' which are clouds of colourful, neutral gas which when whilst cooling down for millions of years began to fuse and form into stars of various sizes from dwarfs to the UY Scuti (the biggest star in the universe currently) (5) explaining how components of galaxies are formed. Galaxies have been studied to have multiple types and characteristics, researched by Mo, H., Van, F. and White, S. (2010 (6) they classify galaxies based on stages of evolution leading to galaxies such as our milky way being a "Barred Spiral Galaxy", however the study by Mo, H and others indicates similarities most galaxies have, one of these being a 'super massive Black hole' at the centre of most galaxies. Images of the milky way having bright centres have a 'super massive black hole in the centre' the idea being that these black holes have a such high gravitational field that it causes stars and entire solar systems to be pulled towards them, maintaining the centripetal forces which is galaxies such as the Milky Way appear circular. The study also theorises that galaxies form in giant nebulas as mentioned previously, with the cooling protons neutrons and electrons beginning to fuse and form giant stars, eventually a giant star that was formed dies out collapses under its own fusion after potentially billions of years and can become a black hole

as researched by () based on images from the Hubble Telescope in the study of supernovae. These Black Holes will eventually grow consuming the energy and matter from these nebulae and grow even larger than typical Black Holes, with the average size of a black holes using 'Schwarzschild's radius' (7) is estimated to be between 30 to 300 km long which is at minimum roughly 5 times earths radius and with, at minimum, being 10 times the weight of our Sun. This forms a core for a 'E0' Galaxy to be formed

Conclusion: Formation:

To conclude Galaxies are formed over millions of years based on the expansion of the universe leading pockets of atomic matter and dark matter to form clouds of hot dense gas that eventually form clusters of stars and planets and when a giant star succumbs to the imbalance of fusion to maintain energy, a supernovae is inevitable which will the sudden outburst of energy causes space to "collapse" in on itself and form a Black Hole, which consumes the matter around it and grows to immense size, with this proportional relationship with Gravitational Field strength causing its gravitational pull to increase the amount of entities surrounding the Black Hole forming an E0 galaxy which will evolve when more stars and planets are moved by the pull of the black hole. This is, based on widespread research from respected organisations, an accurate and logical estimation of the formation of Galaxies.

Reference links:

(1) HubbleSite.org. (n.d.). *GALAXIES*. [online] Available at: <https://hubblesite.org/science/galaxies#section-2b9af6c0-1839-4974-9dde-025337fdbb35>.

(2) LaRocco, C. and Rothstein, B. (n.d.). *THE BIG BANG*. [online] websites.umich.edu. Available at: <https://websites.umich.edu/~gs265/bigbang.htm>.

(3) NASA (2023). *Edwin Hubble - NASA Science*. [online] science.nasa.gov. Available at: <https://science.nasa.gov/people/edwin-hubble/>.

(4) Choi, C.Q. (2017). *Our Expanding Universe: Age, History & Other Facts*. [online] Space.com. Available at: <https://www.space.com/52-the-expanding-universe-from-the-big-bang-to-today.html>.

(5) Testbook. (2024). *Biggest Star in the Universe - Check 12 Cosmic Giants of the Universe*. [online] Available at: <https://testbook.com/static-gk/biggest-star-in-the-universe>

(6) Mo, H., Van, F. and White, S. (2010). *Galaxy formation and evolution*. Cambridge ; New York: Cambridge University Press: <https://www-cambridge-org.surrey.idm.oclc.org/core/books/galaxy-formation-and-evolution/E236D9F26B797202BCA28637BF17E75F>

(7) www.sciencedirect.com. (n.d.). *Schwarzschild Radius - an overview | ScienceDirect Topics*. [online] Available at: <https://www.sciencedirect.com/topics/physics-and-astronomy/schwarzschild-radius>.